Emergency Training Simulator POC

Markdown documentation PDF copy

System Layers

Source: docs/architecture/system-layers.md

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Boundary: this project is still a POC, not operational guidance.

System Layers

This project is split into small layers so each part has one clear job.

Current backend slice

1. API layer receives WebSocket messages and rejects messages that do not match the contract.
2. Session manager keeps actions in order for one session.
3. Scenario engine applies the next state change from scenario data.
4. Validation layer checks POC rules and returns ok, warning, or error.
5. Models and contracts keep events, state, and messages shaped consistently.
6. Logger writes each step to local JSONL for replay and review.
7. Client message layer sends state back in a format any client can read.

The first slice uses in-memory sessions and local files. That is enough for the POC.

Future areas

Unreal rendering, remote networking, sensing, AI-assisted authoring, asset generation, and projection deployment are future work. They should connect through clear contracts instead of owning scenario logic.